

Bunching at a Moving Threshold: Firm Responses to Romania’s Micro-Enterprise Tax, 2014–2025

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Abstract

Romanian companies with revenue below a ceiling set in euro pay 1 to 3 percent of revenue instead of 16 percent of profit. The ceiling moved six times between 2013 and 2026, which gives two things no fixed threshold gives: a counterfactual density at each notch taken from years when no ceiling was there, and repeated reassignment of the same firms between a revenue base and a profit base. Using the accounts of every company, 9.6 million firm-years, we find that reported revenue piles up just below the ceiling, follows it when it moves, and within a fixed euro ceiling tracks its lei value at the previous year-end exchange rate. The pile-up is 2 to 8 percent of nearby firms and 0.06 to 0.30 percent of all firms. Firms moved onto the revenue base by a ceiling increase raise their reported margins, and firms moved off it lower them by 8 to 17 percent of the margin, so the fiscal cost of the regime and the revenue gain from narrowing it are both smaller than static estimates imply.

1 Introduction

Many countries tax small firms on their revenue and larger firms on their profit (Sharma et al., 2025; Kanbur and Keen, 2014; Dharmapala et al., 2011). The size limit between the two regimes is a notch: crossing it switches the tax base and, for a firm whose margin is above the ratio of the two rates, raises the tax bill at once. Responses to such notches

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have been measured at VAT thresholds in the United Kingdom (Liu et al., 2021), Finland (Harju et al., 2019) and Spain (Almunia and López-Rodríguez, 2018), and at turnover-tax and corporate-tax cut-offs in Pakistan (Kleven and Waseem, 2013), Armenia (Asatryan and Peichl, 2016), South Africa (Boonzaaier et al., 2019) and Costa Rica (Bachas and Soto, 2021); the choice between the two bases is the subject of Best et al. (2015). In all of these settings the limit is fixed or moves rarely, so the density that would exist without the notch is extrapolated from the density away from it, usually with a polynomial, and Blomquist et al. (2021) show that the result depends on restrictions the data cannot check. Best and Kleven (2018) compare a location before and after a notch was introduced; Marx (2024) uses panel variation.

Romania’s limit moves. The micro-enterprise regime taxes revenue at 1 to 3 percent for companies whose annual revenue is below a euro ceiling, and the ceiling changed six times: €65,000 until 2015, €100,000 in 2016, €500,000 in 2017, €1,000,000 from 2018 to 2022, €500,000 in 2023 and 2024, €250,000 in 2025 and €100,000 from 2026. That variation does two things. Every location that was a notch in one year was an ordinary point of the distribution in others, which supplies a counterfactual without a fitted curve. And every move reassigned a band of firms between a revenue base and a profit base without any action of their own, which lets us see what the base does to what firms report.

The paper’s results are these. The pile-up in reported revenue follows the ceiling up and down, and within a fixed euro ceiling it sits within a few thousand lei of the ceiling’s value at the previous year-end exchange rate and moves with it; a spike at 4,779,300 lei that moves to 4,869,400 lei the next year is not a rounding habit. Measured against the same quantile in ceiling-free years, the excess is 2 to 8 percent of the firms within 20 percent of the ceiling, robust to windows, control sets and bootstraps, and centred on zero at placebo locations; in aggregate it is 0.06 to 0.30 percent of firms. Bunchers are high-margin firms in professional, administrative, health and information services, and half of them gain from the regime even at the 3 percent rate. When the 2018 increase moved firms onto the revenue base their reported margins rose relative to firms left on profit tax; when the 2023 cut moved them back, margins fell by 1 to 2 points, 8 to 17 percent of the margin. Because 79 percent of the revenue at stake in the regime comes from firms below €500,000, and because reported profit falls when firms move to the profit base, both the cost of the regime and the gain from narrowing it are smaller than static estimates imply. A 2023 rule requiring one employee produced about 20,000 one-employee firms in the year it was enacted and pushed the rest out of the regime.

Section 2 describes the regime, Section 3 the data, Section 4 the estimator, Section 5 the bunching results, Section 6 the response of reported margins to the base and its fiscal implications, Section 7 the employee condition and two further margins. Section 8 concludes.

Table 1: The micro-enterprise regime by year of revenue

Revenue year	Ceiling (EUR)	Rate on revenue	Conditions
2013–2015	65,000	3%	Mandatory (opt-out for new firms with capital $\geq$ €25,000). Consulting and management revenue at most 20% (from 2014). Exceeding the ceiling in-year re-taxed the whole year.
2016	100,000	1% ($\geq$ 2 employees), 2% (1), 3% (none)	Mandatory. In-year exit from the quarter of crossing.
2017	500,000 (from Feb.)	1% ($\geq$ 1 employee), 3% (none)	Mandatory, opt-out for capital $\geq$ 45,000 lei. Consulting cap retained.
2018–2022	1,000,000	1% ($\geq$ 1 employee), 3% (none)	Mandatory, opt-out for capital $\geq$ 45,000 lei with two employees (from April 2018). Consulting cap abolished.
2023	500,000	1%	Optional; exit permanent. At least one employee. Consulting cap 20%. A person holding over 25% may be a shareholder in at most three micro-enterprises. Hospitality firms exempt from the ceiling.
2024	500,000	1% up to €60,000; 3% above or in listed activities	At most one micro-enterprise per person holding over 25%. Ceiling tested on linked enterprises combined. Consulting cap reformulated.
2025	250,000	as in 2024	Consulting cap abolished. Status for 2026 tested on 2025 revenue against €100,000.
2026	100,000	1%	Base changed to accounting turnover including linked enterprises.

2 Institutional background

The micro-enterprise income tax (*impozitul pe veniturile microîntreprinderilor*), Title III of the Fiscal Code, replaces the 16 percent profit tax with a tax on revenue for companies whose revenue in the previous fiscal year was below a ceiling set in euro and that meet side conditions. Table 1 gives both by year of revenue; Appendix A gives the acts.

Three features matter. The first is timing. Under the in-year rule, a micro-enterprise whose cumulated revenue passes the ceiling during year t owes profit tax from that quarter (for 2013 to 2015, for the whole year), the ceiling converted at the central bank (BNR) rate at the close of year $t - 1$. Under the year-end rule, status for year $t + 1$ depends on revenue at 31 December of year t at the rate at the close of year t . When the ceiling is unchanged the two rules put the notch a few thousand lei apart and the in-year rule, at the earlier and lower rate, binds. When a cut has been enacted for the following year a firm faces two notches. That happened in 2022, when Ordinance 16/2022 of July set €500,000 for 2023 status on 2022 revenue and Law 370/2022 of 20 December deferred the revenue test to 2023 revenue; and in 2025, when Ordinance 156/2024 of 31 December 2024 set €250,000 for 2025 and €100,000 for 2026 status on 2025 revenue. A 2016 ceiling of €100,000, enacted in October 2015 and tested on 2015 revenue, is a further year-end

notch that we return to below.

The second feature is the tax base. The ceiling is tested on the taxable base of Article 53, which leaves out items such as subsidies, provision reversals and, from 2021, dividends received. The accounts report total revenue, which is the same or larger; where the excluded items are not zero a firm looks closer to the ceiling than it is, which shifts the bunching region slightly to the right in our data.

The third is the side conditions. The employee condition is met by a full-time contract, part-time contracts adding up to one, or a paid mandate contract; linked enterprises are defined by holdings above 25 percent; the activities taxed at 3 percent regardless of revenue in 2024 and 2025 are software, hospitality, legal and medical services. The consulting cap, in force in 2014–2017 and, reformulated, in 2023–2024, is a margin the accounts do not show. The VAT exemption for small firms is a separate notch in lei, 220,000 lei until March 2018, 300,000 from 1 April 2018 and 395,000 from 1 September 2025, which Appendix C treats.

3 Data

The Ministry of Finance publishes, for each fiscal year, the main figures from the annual accounts of every company, identified by fiscal code (CUI) and CAEN activity code: balance-sheet totals, net turnover, total revenue, total expenses, gross and net profit or loss, and average headcount. The layout is stable from 2016; the 2014 and 2015 files use an older set of columns, which we harmonise. The 2021, 2023 and 2025 files were published a few weeks after the filing deadline and the others are later re-issues, so late filers are covered unevenly. Table 2 summarises the panel: 9.58 million firm-years after removing 32 duplicates, 1.56 million firms. Total revenue is the variable for the micro-enterprise notch.

The accounts do not say which regime a firm is in. We recover it in two ways. Gross profit minus net profit is the income tax accrued; divided by revenue it is roughly 1, 2 or 3 percent for micro-enterprises, and divided by pre-tax profit it is roughly 16 percent for profit-tax firms. The ANAF taxpayer registry, a snapshot dated 31 August 2023, has one column per tax a firm is registered for. Table 3 cross-tabulates the two for 2023: of firms registered for the micro-enterprise tax, 84 percent show a micro pattern and 1 percent a profit-tax pattern; of firms registered for profit tax, 22 percent show the profit-tax pattern and half paid no tax. Where both are decisive, 57 percent of firms, they agree in 92.5 percent of cases. The inference picks out micro-enterprises well and profit-tax firms poorly, so we use the registry flag where membership matters and the accounts-based measure as a check.

Headcount is blank for a quarter of small firms; we treat blanks as zero, and among micro-enterprises three quarters of blank-headcount firms paid the no-employee rate. The

Table 2: Statement panel, 2014–2025

Year	Firm-years	Revenue > 0	Median revenue (k lei)	EUR/RON, 31 Dec
2014	676,868	495,914	124	4.4821
2015	681,821	505,212	141	4.5245
2016	700,249	526,282	146	4.5411
2017	732,051	553,283	151	4.6597
2018	757,929	579,309	162	4.6639
2019	792,376	616,900	173	4.7793
2020	815,779	636,361	162	4.8694
2021	781,470	624,795	188	4.9481
2022	862,564	693,309	207	4.9474
2023	890,836	716,718	223	4.9746
2024	968,594	769,620	226	4.9741
2025	918,232	738,762	239	5.0985

Source: Ministry of Finance annual financial statements (micro-entity and full forms) via data.gov.ro; BNR year-end reference rate. Median revenue is over firms with positive revenue. 32 duplicated firm-years (30 in 2016) removed; about 600 firm-years per year with negative revenue are excluded from all windows.

Table 3: Tax regime inferred from the 2023 statements versus the ANAF tax-vector flag (snapshot of 31 August 2023)

Regime inferred from statements	ANAF: micro	ANAF: profit	ANAF: other/none	Not in ANAF
Micro (tax $\approx$ 1–3% of revenue)	277,265	19,677	312	7,102
Micro (tax > 0, pre-tax loss)	103,544	14,248	87	1,966
Profit tax ($\approx$ 16% of profit)	4,421	92,503	64	2,045
No tax paid	34,486	220,021	273	10,942
Ambiguous	16,047	30,596	60	437
Other	19,272	34,966	158	344

ANAF flags: IMP120 = registered for micro-enterprise income tax, IMP100 = registered for profit tax. Where both measures are decisive, they agree for 92.5% of firms.

field is a whole number and behaves as a flag for having had an employee during the year rather than a time-weighted average, so late hires are counted. The trade register gives each firm’s registration date, address and administrators as of the snapshot of 2 September 2026; persons are keyed on name, birth date and birth place and hashed. Administrators, not shareholders, are our proxy for ownership; 85 percent of the limited-liability companies in the sample have exactly one. Appendix B gives details. Exchange rates are BNR reference rates at the last quotation of each year.

4 Empirical strategy

For each year t the notch location is $L_t = T_t e_{t-1}$, where T_t is the euro ceiling for revenue in year t and e_{t-1} the BNR rate at the close of year $t - 1$. Let F_t be the distribution

of revenue in year t and $q_t = F_t(L_t)$ the quantile of the notch; for every other year c let $L_c^* = F_c^{-1}(q_t)$ be the revenue at the same quantile. Year c is an admissible control if no known notch of year c , the micro ceiling, an enacted or announced future ceiling, the €60,000 rate boundary, or the VAT threshold, lies within 25 percent of L_c^* . Bins are 1 percent of the location wide in a window of ± 20 percent; shares $s_{t,k}$ are bin counts divided by the count in the window, and the counterfactual share is the mean of $s_{c,k}$ over admissible controls. Excess mass is

$$b_t = \frac{\sum_{k \in B} (s_{t,k} - \bar{s}_{c,k})}{\frac{1}{|B|} \sum_{k \in B} \bar{s}_{c,k}}, \quad (1)$$

with B the five bins just below the notch; m_t is the same sum over the five bins just above. With 1 percent bins b is roughly the width of the bunching region in percent of the ceiling. The identifying assumption is that the shape of the revenue density around a given quantile is the same across years apart from the notch. Quantile matching makes the counterfactual immune to inflation and to growth in firm size, not to changes in shape, so we test it: the estimator is run at 77 placebo locations, at 0.45 to 2.5 times each year's ceiling and at least 25 percent from any notch, where the true excess is zero. Because shares are normalised within the window, the excess and missing mass offset by construction; we therefore also compute a count-based version in which each bin is a share of all firms in the year, N_t , so that the integration constraint becomes a test of where bunchers come from. Uncertainty is reported three ways: a multinomial bootstrap of bin counts, a block bootstrap that resamples control years, and the placebo distribution. Appendix Table 20 lists every control year with its matched revenue and distance to the nearest notch; the closest cases are 26 to 34 percent, all for the 2014–2016 notches against VAT thresholds, and tightening the rule to 35 percent changes no estimate from 2017 on.

Three things the estimator does not deliver are stated once here. It does not locate the missing mass, which is spread over the window above the notch rather than concentrated next to it, so the standard conversion of excess mass to an elasticity has no anchor. The tax change on crossing depends on each firm's margin, because the base switches, so a representative-firm elasticity has no clear meaning. And reported margins themselves respond to the base, as Section 6 shows. Appendix Table 24 reports the Kleven–Waseem reduced-form elasticity under two proxies for the bunching width, which differ by three orders of magnitude; we report excess mass and excess shares instead.

Table 4: Location of the revenue spike in lei, 2018–2022

Year	€1M, prev. rate (lei)	Modal 5,000-lei bin	Firms	Bin centre – threshold	Current – prev. rate
2018	4,659,700	4,655,000-4,660,000	59	-2,200	+4,200
2019	4,663,900	4,660,000-4,665,000	73	-1,400	+115,400
2020	4,779,300	4,775,000-4,780,000	95	-1,800	+90,100
2021	4,869,400	4,865,000-4,870,000	105	-1,900	+78,700
2022	4,948,100	4,940,000-4,945,000	101	-5,600	-700

Modal 5,000-lei bin within 60,000 lei of either candidate threshold. The last column gives the lei value of €1M at the current year-end rate minus its value at the previous year-end rate (the dotted line in Figure 1). At 1,000-lei resolution the modal bins hold only 16 to 31 firms and are noisier; they lie 2,000 to 5,000 lei below the previous-rate threshold in 2020–2022 and further below in 2018–2019.

5 Bunching at the ceiling

5.1 The spike follows the ceiling and the exchange rate

Figure 2 shows a spike in the bin just below the ceiling in every year: at €65,000 in 2014 and 2015, €100,000 in 2016, €500,000 in 2017, €1,000,000 from 2018 to 2022, €500,000 in 2023 and 2024 and €250,000 in 2025. Figure 1 looks inside the five years at €1,000,000, when the ceiling’s lei value rose from 4.66 to 4.95 million as the leu depreciated. The spike moves with it each year; Table 4 puts the modal 5,000-lei bin 1,400 to 5,600 lei below the ceiling at the previous year-end rate, the rate the in-year rule prescribes, while the ceiling at the current year-end rate is 79,000 to 115,000 lei further right in 2019 to 2021. Nobody rounds revenue to 4,779,300 lei, and a spike at that value that moves to 4,869,400 lei the next year is not an accounting convention.

5.2 Difference-in-bunching estimates

Table 5 reports the estimates and Figure 3 the densities for four notches; all fourteen are in Appendix Figure 6. In every year the treated density exceeds the same-quantile control density in the bins just below the notch, and the control density is smooth through it. Excess mass in the 5 percent below the notch ranges from 1.9 percent of firms in the window, at the €500,000 ceiling of 2017 and the €250,000 ceiling of 2025, to 7.6 percent at the €500,000 ceiling of 2023. As a share of all filers it is 0.06 to 0.30 percent. Table 6 shows that a block bootstrap over control years gives standard errors of 0.08 to 0.16, that dropping any one control year moves b by at most 0.1, and that narrowing or widening the window moves b without moving the excess counts; Appendix Table 21 adds that control years before and after the treated year agree, that correcting the control quantile for the bunchers changes b by at most 0.1, and that continuing firms give b lower by 0.05 to 0.2.

The placebo distribution is the relevant benchmark. Across the 77 locations b has mean 0.01 and standard deviation 0.15, but its tail is fat: the 95th and 99th percentiles of $|b|$ are 0.29 and 0.42 and the maximum is 0.55. That maximum sits at 466,000 lei in 2015,

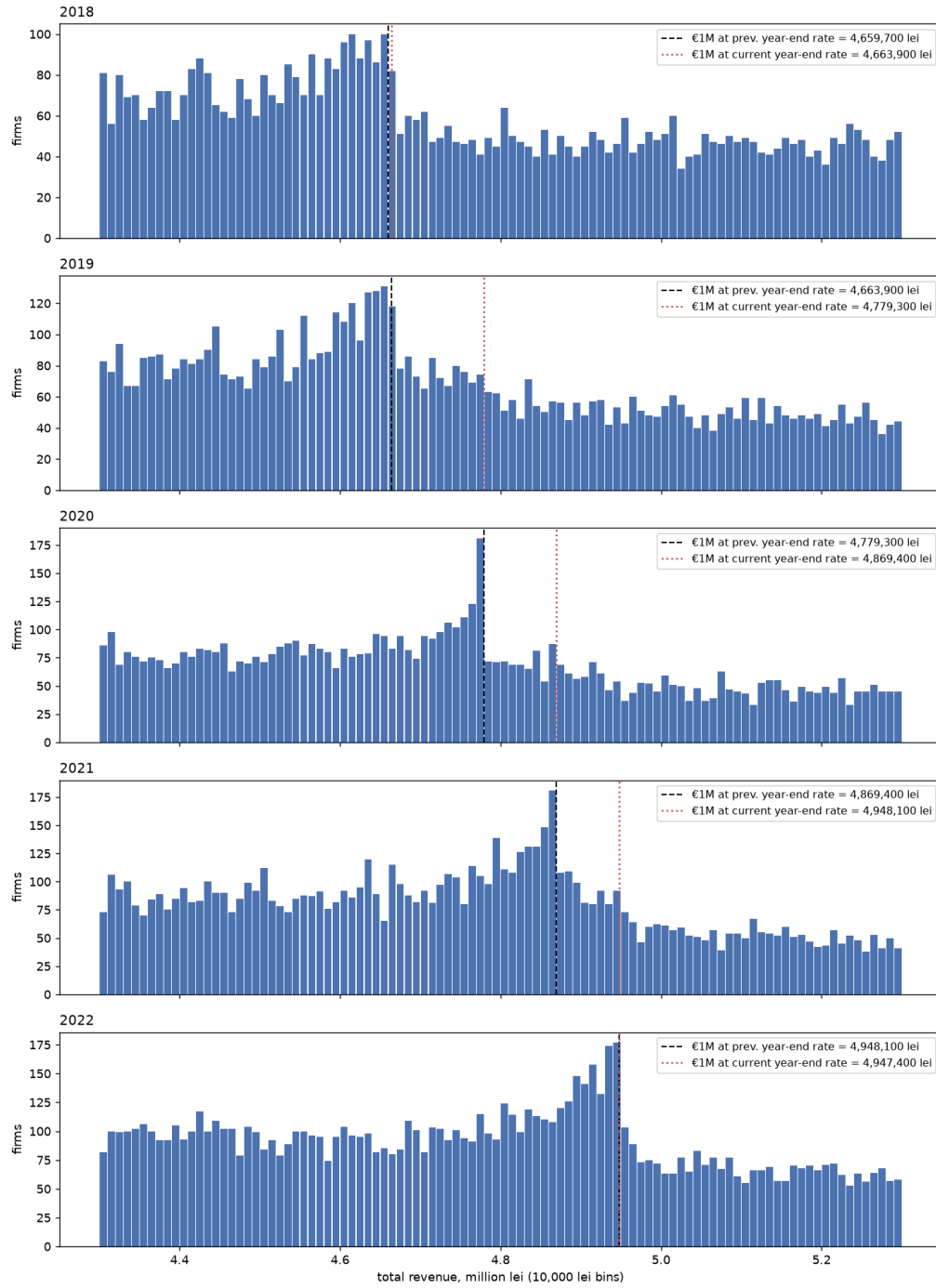


Figure 1: Total revenue in lei around €1,000,000, 2018–2022, 10,000-lei bins. Dashed: ceiling at the previous year-end rate; dotted: at the current year-end rate.

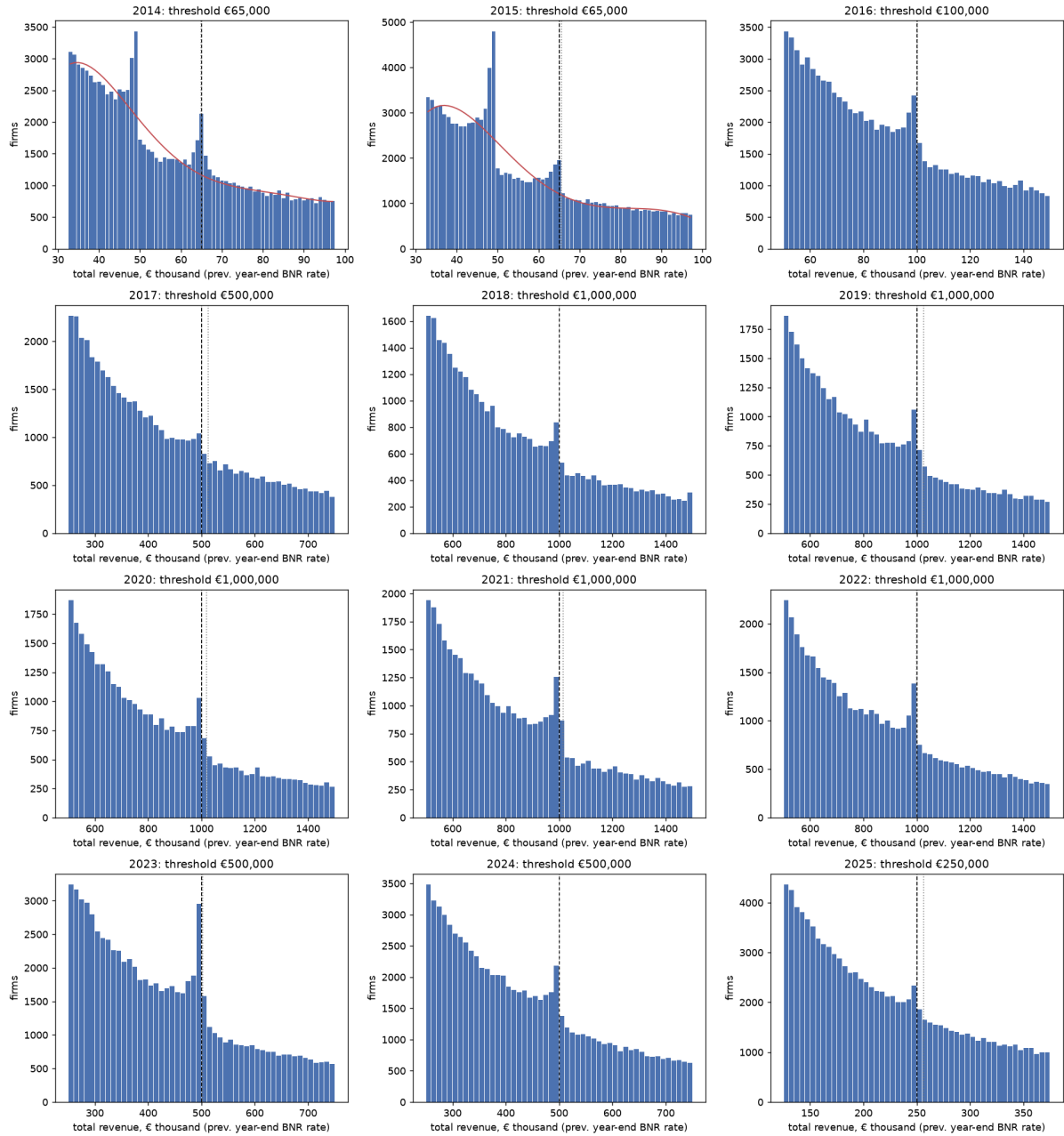


Figure 2: Total revenue around the micro-enterprise ceiling, 2014–2025, converted at the BNR rate at the close of the previous year. Dashed: ceiling; dotted: ceiling at the current year-end rate. In 2014 and 2015 the taller spike at about €49,000 is the VAT threshold of 220,000 lei; the red line there is a polynomial counterfactual, shown because it passes through that spike.

within 3 percent of the €100,000 ceiling for 2016 tested on 2015 revenue, a notch our list had missed; the 2015 distribution has a spike just above that ceiling’s lei value, and the placebo grid landed on it. Excluding locations near it leaves 74 placebos with standard deviation 0.13, 95th percentile 0.27 and maximum 0.39. Every notch estimate exceeds that maximum, most by a factor of two or more; the 2017 and 2025 in-year estimates, 0.76 and 0.73, are the weakest. Two things dilute the 2025 notch: the enacted €100,000 ceiling for 2026 competed for attention, and at the 3 percent rate then in force the break-even

Table 5: Difference-in-bunching estimates

Notch	Location (M lei)	Firms in window	b (SE)	m	Excess firms	% of window	Range over 3–8% regions
2014: in-year, €65,000	0.292	33,859	1.15 (0.08)	0.77	986	2.9	931–986
2015: in-year, €65,000	0.291	34,896	1.71 (0.09)	-0.39	1,526	4.4	1,232–1,674
2016: in-year, €100,000	0.452	32,891	1.56 (0.08)	-0.25	1,321	4.0	985–1,493
2017: in-year, €500,000	2.271	17,172	0.76 (0.11)	-0.26	332	1.9	262–370
2018: in-year, €1,000,000	4.660	11,424	1.41 (0.14)	-0.59	406	3.5	341–467
2019: in-year, €1,000,000	4.664	13,105	1.67 (0.13)	0.05	558	4.3	494–605
2020: in-year, €1,000,000	4.779	12,693	1.75 (0.14)	-0.01	576	4.5	473–655
2021: in-year, €1,000,000	4.869	14,394	2.15 (0.13)	-0.08	793	5.5	670–916
2022: in-year, €1,000,000	4.948	16,448	1.92 (0.13)	-0.45	808	4.9	711–829
2023: in-year, €500,000	2.474	28,379	2.99 (0.11)	-0.14	2,151	7.6	1,850–2,388
2024: in-year, €500,000	2.487	28,619	1.62 (0.10)	-0.28	1,179	4.1	945–1,339
2025: in-year, €250,000	1.244	37,048	0.73 (0.08)	-0.06	690	1.9	588–697
2022: announced, €500,000	2.474	24,241	0.80 (0.10)	-0.13	494	2.0	359–531
2025: year-end, €100,000	0.497	55,232	1.47 (0.07)	0.47	2,059	3.7	1,616–2,278

Location = EUR threshold $\times$ BNR rate at the close of the previous year (in-year notches; the 2025 year-end notch at the end-2024 rate). Window $\pm 20\%$, bins 1% of the location. Control years are listed in Appendix Table 20. b = excess mass in the five bins below the notch in units of counterfactual firms per bin; m = the same over the five bins above, negative meaning a deficit (a hole) and positive an excess; SE from 300 multinomial bootstrap draws. Excess firms = excess in the 5% region below the notch, with its range over 3% and 8% regions.

margin was above the median buncher's.

Table 7 gives the count-based version. It changes the reading in one way. At every large notch the treated year has more firms than the counterfactual across the whole 20 percent below the notch, not only in the five bins next to it, and a deficit in the 20 percent above of at most a seventh of the excess; the net difference over the window is 10 to 17 percent of the window for the 2019 to 2024 notches, against a placebo 95th percentile of 5 percent. Bunchers are therefore not drawn from the region just above the ceiling. The pattern is what a dynamic response produces: firms below the ceiling grow less, so the density below it rises over the years the ceiling is in place, which is also the build-up from 3.5 percent of the window in 2018 to 5.5 percent in 2021 at the €1,000,000 ceiling that Kleven and Waseem (2013) document for Pakistan. It also means the width of the bunching region, and any elasticity derived from it, is not identified in these data.

Two further features of Table 5. The 2023 response is the largest and falls to 4.1 percent in 2024 at the same ceiling; the ceiling halved and exit became permanent that year. The share of 2023 bunchers that had been above €500,000 in 2022, 29.6 percent, is no higher relative to a comparison group than in years when that point was not a ceiling (Appendix Table 22), so it is mean reversion, not displaced firms requalifying. And the two extra rows show a second notch operating: in 2022 a spike of 2.0 percent of the window appears at €500,000, the ceiling firms expected for 2023 status until the December deferral, with none at that location in 2021; in 2025 a spike of 3.7 percent appears at €100,000, the ceiling enacted for 2026. That spike runs from 494,000 to 500,000 lei at 1,000-lei resolution, straddling the €100,000 value at the end-2024 rate, 497,410 lei, which is why m is positive for that row; the data cannot separate targeting of the last observable rate from targeting of 500,000 lei as a round number. For comparison, a polynomial counterfactual of the kind used by Chetty et al. (2011) gives 0.4 to 2.0 at baseline and, across 27 combinations

Table 6: Robustness of the difference-in-bunching estimates

Notch	b	Block-bootstrap SE	Window $\pm 10\%$	Window $\pm 30\%$	Leave-one-out range	B/N_t (%)
2014: in-year, €65,000	1.15	0.08	0.74	0.78	1.13–1.18	0.20
2015: in-year, €65,000	1.71	0.08	1.36	1.55	1.69–1.72	0.30
2016: in-year, €100,000	1.56	0.11	1.23	1.70	1.54–1.64	0.25
2017: in-year, €500,000	0.76	0.12	0.62	0.79	0.73–0.79	0.06
2018: in-year, €1,000,000	1.41	0.14	1.20	1.46	1.38–1.43	0.07
2019: in-year, €1,000,000	1.67	0.14	1.27	1.87	1.63–1.71	0.09
2020: in-year, €1,000,000	1.75	0.16	1.35	1.93	1.71–1.83	0.09
2021: in-year, €1,000,000	2.15	0.15	1.59	2.39	2.10–2.20	0.13
2022: in-year, €1,000,000	1.92	0.14	1.58	2.15	1.87–1.97	0.12
2023: in-year, €500,000	2.99	0.11	2.19	3.35	2.97–3.02	0.30
2024: in-year, €500,000	1.62	0.09	1.26	1.78	1.58–1.64	0.15
2025: in-year, €250,000	0.73	0.08	0.57	0.78	0.71–0.75	0.09
2022: announced, €500,000	0.80	0.10	0.60	0.89	0.78–0.82	0.07
2025: year-end, €100,000	1.47	0.07	1.02	1.11	1.44–1.50	0.28

Baseline: window $\pm 20\%$, all admissible control years. Block-bootstrap SE: control years resampled with replacement and treated bin counts resampled, 500 draws. Leave-one-out: range of b dropping one control year at a time. B/N_t : excess firms in the 5% region as a share of all filers with positive revenue in year t . Further checks in Appendix Table 21.

Table 7: Count-based difference-in-bunching: firms relative to the same-quantile counterfactual scaled to the year’s filers

Notch	b , count-based	Excess, 5 bins below	5 bins above	5–20% above	5–20% below	Net, whole window	Net, % of window
2014: in-year, €65,000	1.23	1,043	+680	+41	-620	+464	1.4
2015: in-year, €65,000	1.75	1,554	-301	-1,446	+120	+228	0.7
2016: in-year, €100,000	1.89	1,523	-12	-730	+769	+1,561	4.7
2017: in-year, €500,000	0.94	399	-44	-119	+245	+525	3.1
2018: in-year, €1,000,000	1.74	477	-93	-212	+300	+565	4.9
2019: in-year, €1,000,000	2.46	736	+176	-66	+721	+1,390	10.6
2020: in-year, €1,000,000	2.50	739	+143	-104	+627	+1,262	9.9
2021: in-year, €1,000,000	3.35	1,057	+213	-2	+1,007	+2,062	14.3
2022: in-year, €1,000,000	2.81	1,049	+48	-29	+861	+1,882	11.4
2023: in-year, €500,000	4.59	2,749	+460	-89	+2,064	+4,723	16.6
2024: in-year, €500,000	2.38	1,551	+157	-42	+1,425	+2,934	10.3
2025: in-year, €250,000	1.00	907	+152	+39	+762	+1,708	4.6
2022: announced, €500,000	0.97	580	+5	-303	+399	+676	2.8
2025: year-end, €100,000	2.29	2,850	+1,363	+649	+2,752	+6,250	11.3

Each bin is a share of all firms with positive revenue in the year rather than of the window, so the counterfactual is not forced to integrate to the treated total. Columns give treated minus counterfactual in firms. At 74 placebo locations the net window difference has mean -0.5% of the window, standard deviation 3.1 and 95th percentile of the absolute value 5.4; the count-based b has standard deviation 0.19 and maximum 0.61.

of bin width, degree and excluded window, anything from negative values to more than twice the baseline.

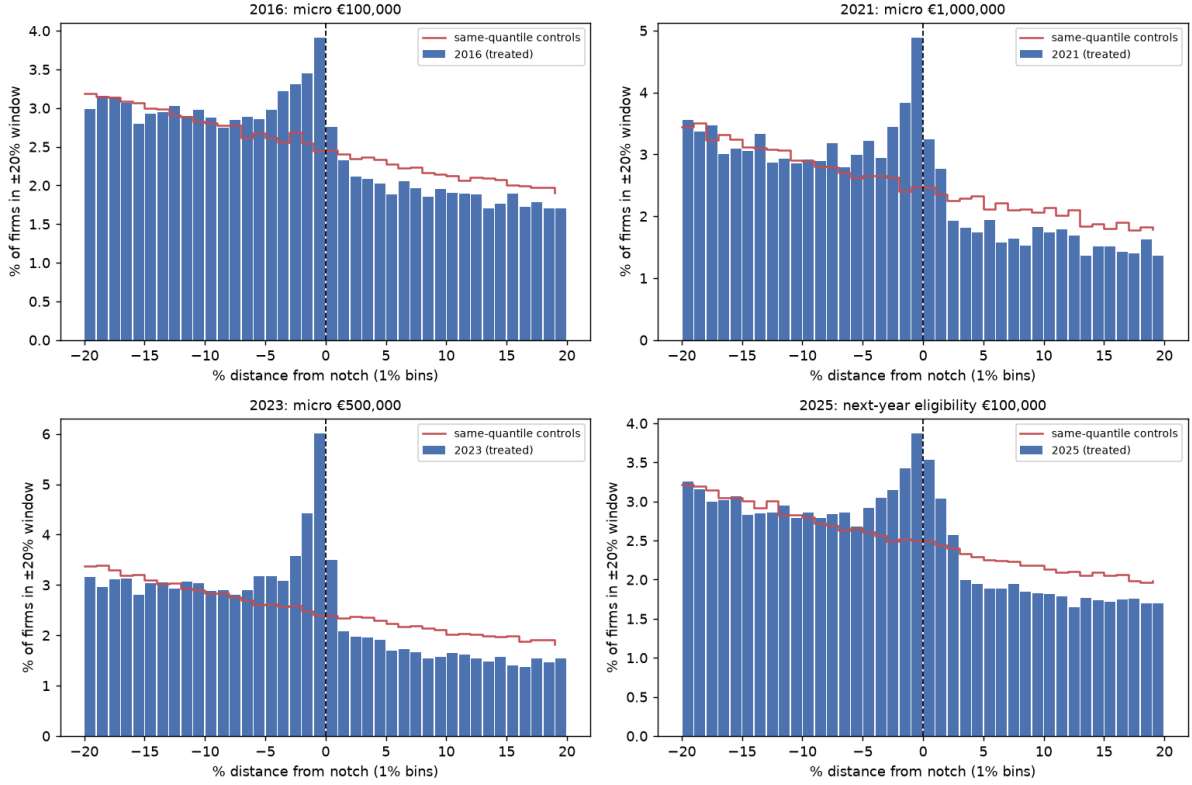


Figure 3: Treated-year density (bars) and same-quantile control density (line) around four notches, bins of 1 percent of the notch location.

5.3 Who bunches

The notch raises taxes only for firms whose margin exceeds the ratio of the two rates, 6.25 percent at a 1 percent revenue tax and 18.75 percent at 3 percent. Table 8 shows that firms in the 5 percent below the ceiling report median margins of 9 to 20 percent against 3 to 10 percent for firms just above, and Table 9 shows where they are: in 2023, excess mass is 11 to 15 percent of the window in professional, administrative, health and real-estate services and in construction, and 10 percent in information services, against 5 percent in trade, 3.5 in transport and 1 in agriculture, and bunchers in the service sectors report margins of 29 to 43 percent.

The 3 percent years pose a puzzle that Table 8 also resolves. In 2014, 2015, 2024 and 2025 the median buncher's margin, 13 to 17 percent, is below the 18.75 percent at which a 3 percent revenue tax saves tax, yet 2014 and 2015 show excess mass of 1.2 and 1.7, more than most 1 percent years. The median hides a split: 47 to 48 percent of bunchers in 2014 and 2015 report margins above 18.75 percent, against 18 to 27 percent of firms just above, and the pattern repeats in 2024 and 2025 with 43 to 45 percent against 28 to 33. Half the bunchers gain from the regime and the other half do not, and both halves sit at the ceiling. For the second half the explanation has to be something other than the tax wedge, and the candidate is the one Harju et al. (2019) find at the Finnish VAT threshold,

Table 8: Pre-tax profit margins by position relative to the ceiling, and the rate bunchers paid

Year	Median margin, bunchers	just above	Bunchers: margin > 6.25% (%)	Bunchers: margin > 18.75% (%)	Just above: > 18.75% (%)	Median rate paid by micro bunchers (%)
2014	0.160	0.027	62	47	18	3.0
2015	0.169	0.058	65	48	27	3.0
2016	0.134	0.053	63	43	24	1.0
2017	0.089	0.062	58	30	21	1.0
2018	0.106	0.057	62	31	19	1.0
2019	0.117	0.070	65	35	22	1.0
2020	0.132	0.078	69	38	22	0.9
2021	0.160	0.097	74	45	31	0.9
2022	0.166	0.101	75	46	31	0.9
2023	0.200	0.098	76	52	31	0.9
2024	0.151	0.081	68	45	28	2.9
2025	0.135	0.093	64	43	33	2.7

Bunchers: 0–5% below the ceiling; just above: 0–10% above. Break-even margins are 6.25% at a 1% revenue tax and 18.75% at 3%. Last column: median of accrued tax over revenue among bunchers inferred to be on the revenue tax; in 2016 the rate depended on headcount and the median buncher paid 1%.

Table 9: Excess mass by sector, in percent of firms within 20% of the notch

CAEN section	2021 (€1M)	2023 (€500k)	2025 (€250k)	Firms in 2023 window	Median margin of 2023 bunchers
Professional services	–	14.8	4.6	2,068	0.39
Administrative services	–	13.3	3.4	1,089	0.29
Construction	7.4	12.6	3.1	4,078	0.25
Health	–	12.0	2.6	896	0.32
Real estate	–	11.2	3.5	908	0.43
Information and communication	–	10.1	5.1	1,124	0.34
Manufacturing	2.9	6.5	1.9	3,008	0.16
Trade	5.3	4.8	0.3	8,426	0.12
Transport	6.1	3.5	-0.2	2,428	0.10
Agriculture	0.5	1.0	1.0	1,653	0.08
Hospitality	–	0.9	0.4	1,571	0.14
Arts, other services	–	–	1.7	–	–

Difference-in-bunching within each CAEN section, matched at the section’s own quantile, same control years as the pooled estimate; sections with fewer than 800 firms in the window omitted. Hospitality faced no revenue ceiling in 2023.

compliance cost: crossing the ceiling meant, in those years, leaving a mandatory regime with a one-line tax base for quarterly profit-tax returns and full cost accounting. In 2016 the rate depended on headcount and the median buncher paid 1 percent, so 2016 is not a 3 percent year.

6 The base changes what firms report

A micro-enterprise’s tax does not depend on its reported profit; a profit-tax firm’s does. Firms on profit tax therefore have a reason to report lower margins, by booking costs aggressively, or by paying the owner a salary, which is deductible under profit tax and taxed on the owner, rather than a dividend, which is not deductible and was taxed at 5 to 8 percent. Both responses lower reported profit; only the first is misreporting. The accounts do not separate personnel costs, so we cannot tell them apart and say so; what we can do is measure the total.

Table 10: First stage and per-switch effects for the margin design

Year	Micro share, treated (%)	Micro share, control (%)	First stage (pp)	DiD, median margin (pp)	Per regime switch (pp)	n	T / C
<i>2018 entry, control band $(1.0, 1.2] \times \text{€}1M$</i>							
2015	14.2	12.8	+0.6	+0.09	–	5,353	3,760
2016	13.8	13.6	–0.6	+0.16	–	5,353	3,760
2017	15.3	14.5	+0.0	+0.00	–	5,353	3,760
2018	55.5	29.2	+25.5	+0.71	+2.78	5,353	3,760
2019	45.3	28.8	+15.8	+0.51	+3.21	5,353	3,760
2020	44.1	32.8	+10.5	+0.48	+4.57	5,353	3,760
<i>2018 entry, control band $(1.2, 1.5] \times \text{€}1M$</i>							
2015	14.2	13.5	–0.1	–0.09	–	5,353	3,807
2016	13.8	12.8	+0.2	–0.26	–	5,353	3,807
2017	15.3	14.5	+0.0	+0.00	–	5,353	3,807
2018	55.5	17.8	+36.9	+0.60	+1.64	5,353	3,807
2019	45.3	21.1	+23.4	+0.34	+1.45	5,353	3,807
2020	44.1	25.4	+18.0	+0.52	+2.90	5,353	3,807
<i>2023 exit, control band $(1.0, 1.2] \times \text{€}1M$</i>							
2020	69.1	58.9	–14.4	–0.06	+0.44	7,556	4,766
2021	70.2	57.3	–11.8	+0.07	–0.62	7,556	4,766
2022	66.3	41.7	+0.0	+0.00	–	7,556	4,766
2023	40.1	24.9	–9.4	–0.95	+10.06	7,556	4,766
2024	19.3	18.5	–23.8	–1.62	+6.79	7,556	4,766
2025	17.5	16.8	–24.0	–1.98	+8.26	7,556	4,766
<i>2023 exit, control band $(1.2, 1.5] \times \text{€}1M$</i>							
2020	69.1	45.9	–18.0	–0.78	+4.36	7,556	4,797
2021	70.2	40.3	–11.3	–0.45	+3.97	7,556	4,797
2022	66.3	25.1	+0.0	+0.00	–	7,556	4,797
2023	40.1	21.4	–22.5	–2.00	+8.92	7,556	4,797
2024	19.3	17.6	–39.5	–2.78	+7.02	7,556	4,797
2025	17.5	15.9	–39.6	–2.81	+7.09	7,556	4,797

Micro share = share of the band inferred to be on the revenue tax that year. First stage = change in the treated band’s micro share minus the control band’s, relative to the base year. Per switch = DiD divided by the first stage, a ratio of medians to means that should be read as an order of magnitude. Bands on two-year average revenue: treated $(0.8, 1.0] \times \text{€}1M$ throughout.

6.1 Design

In 2018 the ceiling rose from €500,000 to €1,000,000, so firms with 2017 revenue between those values were moved from profit tax to revenue tax without doing anything, while firms above €1,000,000 stayed on profit tax. In 2023 the cut reversed the move. Wide bands, $(\text{€}500,000, \text{€}1,000,000]$ against $(\text{€}1,000,000, \text{€}2,000,000]$, compare firms that differ twofold in size and produce mean-reversion drift of up to 1.5 points in a placebo (Appendix Table 23). We therefore use bands 20 percent either side of the cut-off, assigned on two-year average revenue, in balanced panels, and placebos built at the same time with the same bands at €1,500,000 and €2,000,000, where no regime changed. Table 10 gives the first stage. In 2018 the treated band’s micro share jumps from 15 to 55 percent; the control band just above the new ceiling goes from 15 to 29 percent, because some of those firms bunched under it, so the control is partly treated, which is why the entry effect fades over the horizon in the baseline. A control band of 1.2 to 1.5 times the cut-off stays at 15 to 25 percent. In 2023 the treated band’s micro share falls from 66 to 19 percent by 2024, and the control band also loses the regime, so the design is fuzzy on both sides; the wider control band, at 25 percent micro in 2022, is the cleaner one.

Table 11: Reported margins around regime switches: narrowed bands with contemporaneous placebos

Year	Horizon	DiD, median margin, pp (SE)	Placebo at €1.5M	Placebo at €2M	Corrected, pp (SE)
<i>2018 entry: bands on 2016–2017 average revenue, cut-off €1M; placebos at €1.5M and €2M</i>					
2015	-2	+0.09 (0.20)	+0.25	-0.06	-0.01 (0.28)
2016	-1	+0.16 (0.20)	+0.46	-0.04	-0.05 (0.27)
2017	+0	0.00	+0.00	+0.00	0.00
2018	+1	+0.71 (0.21)	+0.28	-0.11	+0.62 (0.26)
2019	+2	+0.51 (0.23)	+0.44	+0.03	+0.27 (0.29)
2020	+3	+0.48 (0.23)	+0.14	-0.08	+0.45 (0.28)
<i>2023 exit: bands on 2021–2022 average revenue, cut-off €1M; placebos at €1.5M and €2M</i>					
2020	-2	-0.06 (0.32)	+0.49	+0.04	-0.33 (0.41)
2021	-1	+0.07 (0.27)	+0.72	+0.21	-0.39 (0.32)
2022	+0	0.00	+0.00	+0.00	0.00
2023	+1	-0.95 (0.26)	+0.09	-0.36	-0.81 (0.33)
2024	+2	-1.62 (0.33)	-0.07	-0.20	-1.48 (0.38)
2025	+3	-1.98 (0.27)	-0.30	+0.06	-1.86 (0.34)

Balanced panels. Treated band: two-year average revenue in $(0.8, 1.0] \times$ the cut-off; control band: $(1.0, 1.2] \times$ the cut-off; DiD = change in the treated band’s median pre-tax margin minus the change in the control band’s, relative to the base year. Placebos: the same construction at cut-offs of €1.5M and €2M in the same base year, where no regime changed. Corrected = DiD minus the mean of the two placebos. SE from 200 bootstrap draws over firms. Firms: 5,353/3,760 treated/control in 2018, 7,556/4,766 in 2023; placebo designs 2,200–4,800 per band.

6.2 Results

Table 11 reports the difference in differences on the median margin. Pre-period differences are within 0.2 points of zero and the contemporaneous placebos within 0.5, so the correction is small and creates no pre-trend. Entry in 2018 raises the treated band’s median margin by 0.7 points in the first year (standard error 0.2) from a base of 5.3 percent, 0.6 (0.3) net of placebos; it is significant only at that horizon, at 0.3 (0.3) and 0.5 (0.3) after, which the contaminated control explains in part. Exit in 2023 lowers it by 1.0, 1.6 and 2.0 points over three years from a base of 11.5 percent, -0.8 (0.3), -1.5 (0.4) and -1.9 (0.3) net of placebos, and by 2.0 to 2.8 points against the wider control band. Per regime switch, dividing the effect by the first stage, the entry effect is 1.6 to 2.8 points and the exit effect 7 to 10 points, a ratio of medians to means that should be read as an order of magnitude; on the base margins those are 30 to 50 percent and 60 to 90 percent of the margin. The asymmetry makes sense: it is easier to start reporting costs against a profit tax than to unwind arrangements on leaving it, and the exit effect builds over years.

6.3 Fiscal magnitudes

Table 12 puts the regime in lei for 2023. Micro-enterprises paid 8.0 billion lei of revenue tax; 16 percent of their reported pre-tax profit would have been 15.5 billion, a static gap of 7.5 billion lei. Of that gap, 79 percent comes from firms with revenue below €500,000 and only 8 percent from the €500,000 to €1,000,000 band where the 2023 bunchers and the reclassified firms sit. The gap is an upper bound, and the previous subsection says by

Table 12: Fiscal magnitudes in 2023, million lei: micro-enterprise tax paid against 16 percent of reported pre-tax profit, by revenue band

Revenue band (€, prev. rate)	Firms	Micro firms	Micro tax paid	16% of their profit	Gap	Gap share (%)	Profit-tax firms	Profit tax paid
< 60k	420,646	216,058	304	1,189	886	12	76,560	417
60-250k	164,407	135,902	782	3,398	2,616	35	7,720	253
250-500k	55,483	45,218	739	3,121	2,382	32	1,970	113
500k-1M	31,385	16,228	616	1,242	626	8	2,896	244
> 1M	44,797	10,343	5,606	6,566	960	13	9,784	6,134

Micro firms are those inferred to be on the revenue tax from the statements; profit at 16% uses reported pre-tax profit floored at zero and is an upper bound, since Section 6 shows reported margins fall by 8 to 17 percent of their level when firms move to profit tax. The gap is the static revenue difference; the share column is each band’s share of the total gap.

how much: firms moved to profit tax in 2023 reported margins 8 to 17 percent lower than their base, so a static simulation that applies 16 percent to the profit those firms reported as micro-enterprises overstates the revenue from reclassifying them by that share. Ban and Buciu (2023) simulate revenue across ceilings from zero to €1,000,000 under exactly that assumption; the correction is first order for the reclassified band and second order for the regime as a whole, because the intramarginal firms, not the threshold, carry the cost.

7 The employee condition and other margins

7.1 The employee condition

From 1 January 2023 a micro-enterprise must have at least one employee, tested at 31 December 2022; the rule was enacted in July 2022. Table 13 shows the share of small firms with exactly one employee rising from 31.9 percent in 2021 to 37.6 in 2022 and 40.7 in 2023. Table 14 follows zero-employee firms: 22.3 percent of them moved to exactly one employee in 2022, against 11.1 to 11.3 percent in the two preceding transitions, or 20.3 percent when the sample is limited to firms present since 2020; among zero-employee firms above €1,500,000, untouched by the rule, the rate stayed at 5 to 6 percent. That is about 20,000 additional one-employee firms. Of micro-enterprises still without employees at the end of 2022, 82 percent were registered for profit tax in 2023 and 15 percent had an employee by the end of 2023. The hiring came in the enactment year and the rest left the regime. The implied tax rates of the 2022 hirers suggest many hired late in the year, but rate inference is noisy for one-employee firms, so we do not lean on the timing. Headcount counts a part-time contract as one, so we read this as compliance at the lowest cost rather than job creation.

7.2 Two further margins, in brief

The same population bunches at the VAT exemption threshold, set in lei, with polynomial excess mass of 2.8 to 6.6 in every year, and when the threshold moved from 300,000 to

Table 13: Employee counts among firms with revenue below €500,000 (% of firms)

Year	0 employees	1	2	5+	Firms
2019	33.6	28.7	13.9	11.8	555,050
2020	33.4	30.6	13.4	11.2	575,855
2021	35.1	31.9	12.1	10.2	558,724
2022	31.1	37.6	11.9	9.1	616,503
2023	28.7	40.7	12.5	8.3	640,536
2024	30.6	40.8	12.2	7.3	686,605
2025	32.6	40.5	11.8	6.4	655,235

Average annual headcount as reported in the statements.

Table 14: Transitions of zero-employee firms (revenue below €500,000) into the next year, conditional on filing in both years (%)

Year t	Firms with 0 employees in t filing in $t + 1$	1 employee in $t + 1$	2+ in $t + 1$	Still 0 in $t + 1$
2019	173,878	11.1	4.7	84.1
2020	165,103	11.3	3.5	85.2
2021	178,783	22.3	3.3	74.4
2022	168,742	14.8	2.9	82.3
2023	169,005	9.4	2.9	87.7
2024	178,415	8.1	2.3	89.6

The one-employee condition was enacted in July 2022 and tested at 31 December 2022. Filing coverage differs across years because some annual files are early uploads, so shares are conditional on a statement being present in $t + 1$.

395,000 lei on 1 September 2025 the spike relocated within the same year. Its move also uncovers the €60,000 boundary between the 1 and 3 percent rates, a two-point notch on the whole base, which draws a spike a fifth the size of the VAT spike. Appendix C gives the tables and figures. Firms in the 5 percent below the ceiling are more likely than equally sized and aged firms just above it to have an administrator who registered another firm in the previous eighteen months, 2.7 against 2.0 percent, or a new firm at their address, 12.2 against 9.2 percent; but the combined revenue of firms sharing an administrator does not bunch at the ceiling in 2024, when the law began to test it that way, so we report splitting as a correlation rather than a mechanism. Appendix D gives the tables.

8 Conclusion

A ceiling that moved six times gives a counterfactual density without a fitted curve and reassigns the same firms between a revenue base and a profit base. The first shows that bunching at the ceiling is real, follows the ceiling and the exchange rate, and amounts to 2 to 8 percent of nearby firms and 0.06 to 0.30 percent of all firms, with no deficit just above the ceiling, so the response is a slow accumulation below it rather than a jump from just above. The second shows that reported margins rise when firms are moved onto the revenue base and fall by 8 to 17 percent of the margin when they are moved off it. Together they say that both the fiscal cost of the regime and the revenue from narrowing it are smaller than static estimates imply: 79 percent of the cost sits with firms well below the ceiling, and the profit those firms would report under profit tax is lower than the profit they report now.

Three design points follow, limited to the margins we measure. The ceiling is the wrong lever: moving it changes the behaviour of a few thousand firms and the bills of hundreds of thousands. The regime’s advantage is concentrated in high-margin services, and a structure that ties the rate to the margin, the minimum-tax design that Sharma et al. (2025) show dominates threshold systems, would remove both the notch and the incentive to report margins differently on its two sides. Side conditions that can be met with a single contract produce that contract, as the employee rule did. What the data do not settle is how much of the margin response is misreporting and how much is income shifting between salary and dividends; that needs personnel costs, which the public accounts do not carry.

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A Legal timeline

The regime’s parameters were changed by Ordinance 8/2013 (mandatory regime, €65,000, 3 percent, from 1 February 2013), Ordinance 102/2013 (consulting cap, 2014), Ordinance 50/2015 of October 2015 (€100,000 and employee-dependent rates, 2016, tested on 2015 revenue), Ordinance 3/2017 (€500,000 from 1 February 2017), Ordinance 79/2017 (€1,000,000 from 2018, consulting cap abolished), Ordinance 25/2018 (capital and employee opt-out from April 2018), Ordinance 16/2022 of 15 July 2022 (€500,000, employee condition, three-firm rule, optional regime with permanent exit, from 2023) as amended by Law 370/2022 of 20 December 2022 (revenue test deferred to 2023 revenue), Law 296/2023 (€60,000 rate boundary and activity list, 2024), Ordinance 115/2023 (one-firm rule and linked-enterprise aggregation, 2024), Ordinance 156/2024 of 31 December 2024 (€250,000 for 2025, €100,000 for 2026, consulting cap abolished), Ordinance 89/2025 (flat 1 percent from 2026) and Ordinance 8/2026 (turnover base from 2026). Exchange-rate provisions: Article 47(1)(c) for the year-end test and Article 52 for the in-year test of Law 227/2015, and before 2016 Article 112⁶ of Law 571/2003 with its implementing norms; in every version the in-year test uses the BNR rate at the close of the previous financial year. VAT thresholds: Article 310 of Law 227/2015 as amended by Law 72/2018 (300,000 lei from 1 April 2018) and Ordinance 22/2025 (395,000 lei from 1 September 2025). A full table with Monitorul Oficial references accompanies the replication files. Policy assessments of the regime are in IMF (2025), Consiliul Fiscal (2024), Ban and Buciu (2023) and Lazăr (2024).

B Data details

Regime inference. The gap between gross and net profit also contains the specific tax on hospitality firms in 2017–2022 and other taxes on the result, net of sponsorship and cash-register credits, so ratios such as 0.8 or 2.4 percent occur and a firm with positive tax and a pre-tax loss is most likely, but not certainly, a micro-enterprise. Of firms registered for profit tax, 8 percent show a micro pattern and half paid no tax, which is what loss-making and dormant profit-tax firms look like; 2.6 percent of 2023 filers are not in the registry. In the registry’s column dictionary, obligation 120 is the micro-enterprise income tax and obligation 100 the profit tax.

Employees. Headcount is reported as zero for a further 5 percent of small firms beyond the blanks. Of firms that go from zero to one employee in 2022, 12 percent paid the no-employee rate for most of that year, which a true average would have rounded to zero; late-year hires are therefore counted. The legal test is met by a full-time contract, by part-time contracts adding up to one, or by a paid mandate contract, none of which the

field distinguishes.

Trade register. The register lists administrators as of the snapshot date, not shareholders and not past administrators; the open data has a representatives table only in snapshots from November 2025 onwards, so administrators at the time of the 2018–2023 observations cannot be recovered. Company names carry no usable marker of sole ownership. Nearly all filers, 96 to 99 percent by year, link to at least one administrator, and the linked share is the same across the groups compared. Four percent of administrator records lack a birth date, so their key is the name alone; results are reported with and without them. Names, birth dates and addresses are used only to build hashed keys and appear in no output or replication file. Net turnover, used for the VAT threshold, is blank for a fifth of micro-entity forms but for under 1 percent of firms near that threshold.

C The VAT threshold and the rate boundary

The VAT threshold moved from 300,000 to 395,000 lei on 1 September 2025. In the 2025 distribution of turnover the spike at 300,000 lei is gone and a spike at 395,000 has replaced it, with polynomial excess mass of 3.0 against 0.6 at that location in 2024 (Table 15); part of the spike is mechanical, since the ordinance let firms that crossed 300,000 lei in August 2025 stay unregistered up to the new threshold. The difference-in-bunching estimator is not available at this threshold because a VAT notch exists in every year within 25 percent of the same-quantile location. The polynomial estimates in Table 15 and Figure 5 show excess mass between 2.8 and 6.6 in every year from 2014 to 2024 at bin widths of 0.7 to 0.9 percent of the threshold, against 0.4 to 2.0 for the micro-enterprise notch at bin widths of 1.5 to 2 percent. The legal base is the turnover of Article 310(2), which differs from net turnover by exempt and ancillary items. In 2024 the €60,000 rate boundary and the VAT threshold are 1,500 lei apart and the combined spike has polynomial excess mass of 4.9; in 2025, with the VAT threshold at 395,000 lei, a spike remains at 296,000 to 298,000 lei, the lei value of €60,000 at the end-2024 rate, with excess mass of 0.9 (Table 16, Figure 4). Firms respond to the rate boundary, but weakly; the activity list already puts software, legal and medical firms on 3 percent regardless of revenue, and those are the high-margin firms for which two points of revenue matter most.

Table 15: Bunching of net turnover at the VAT registration threshold

Year	VAT threshold (lei)	Firms in window	b (polynomial)
2014	220,000	83,751	2.84
2015	220,000	91,256	4.21
2016	220,000	97,622	4.76
2017	220,000	105,826	5.39
2018	300,000	102,168	3.64
2019	300,000	113,716	5.09
2020	300,000	111,913	4.90
2021	300,000	117,918	5.78
2022	300,000	135,830	6.61
2023	300,000	145,471	6.62
2024	300,000	157,977	6.09
2025	300,000	157,676	0.79
2025	395,000	147,085	3.03

Polynomial counterfactual (degree 5, three bins excluded on each side), 2,000-lei bins, window $\pm 40\%$. The threshold rose to 395,000 lei on 1 September 2025; the two 2025 rows test the old and the new location (at 395,000 lei the 2024 placebo gives 0.62).

Table 16: The 1%/3% rate boundary at €60,000 and the VAT threshold, 2023–2025

Year	€60,000 in lei	VAT threshold (lei)	b at €60,000	b at 300,000	Modal 2,000-lei bin, 250–450k	Firms in modal / median bin
2023	296,844	300,000	3.17	5.89	298,000-300,000	5,215 / 742
2024	298,476	300,000	4.88	5.53	298,000-300,000	5,121 / 801
2025	298,446	300,000 (to Aug. 2025)	0.94	0.92	296,000-298,000	1,910 / 1,101

Polynomial excess mass, 1,500-lei bins, window $\pm 30\%$. In 2023 no rate boundary existed; in 2024 the boundary and the VAT threshold lie 1,500 lei apart; in 2025 the VAT threshold moved to 395,000 lei on 1 September.

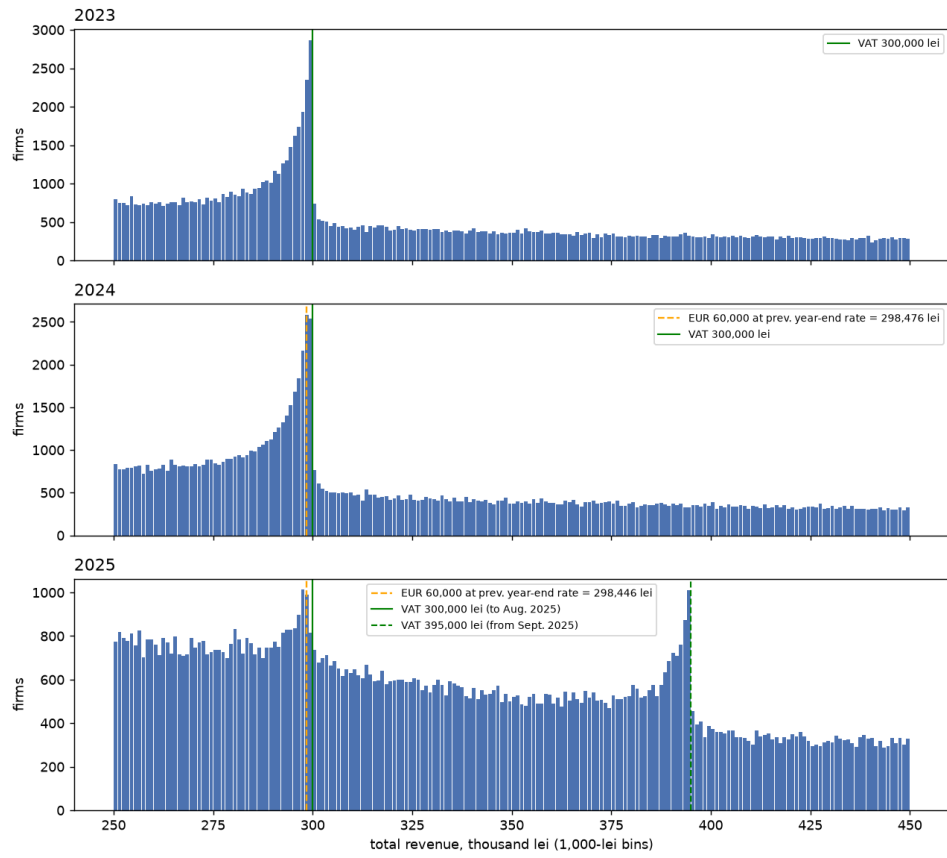


Figure 4: Total revenue from 250,000 to 450,000 lei, 2023–2025, 1,000-lei bins, with the €60,000 rate boundary and the VAT thresholds.

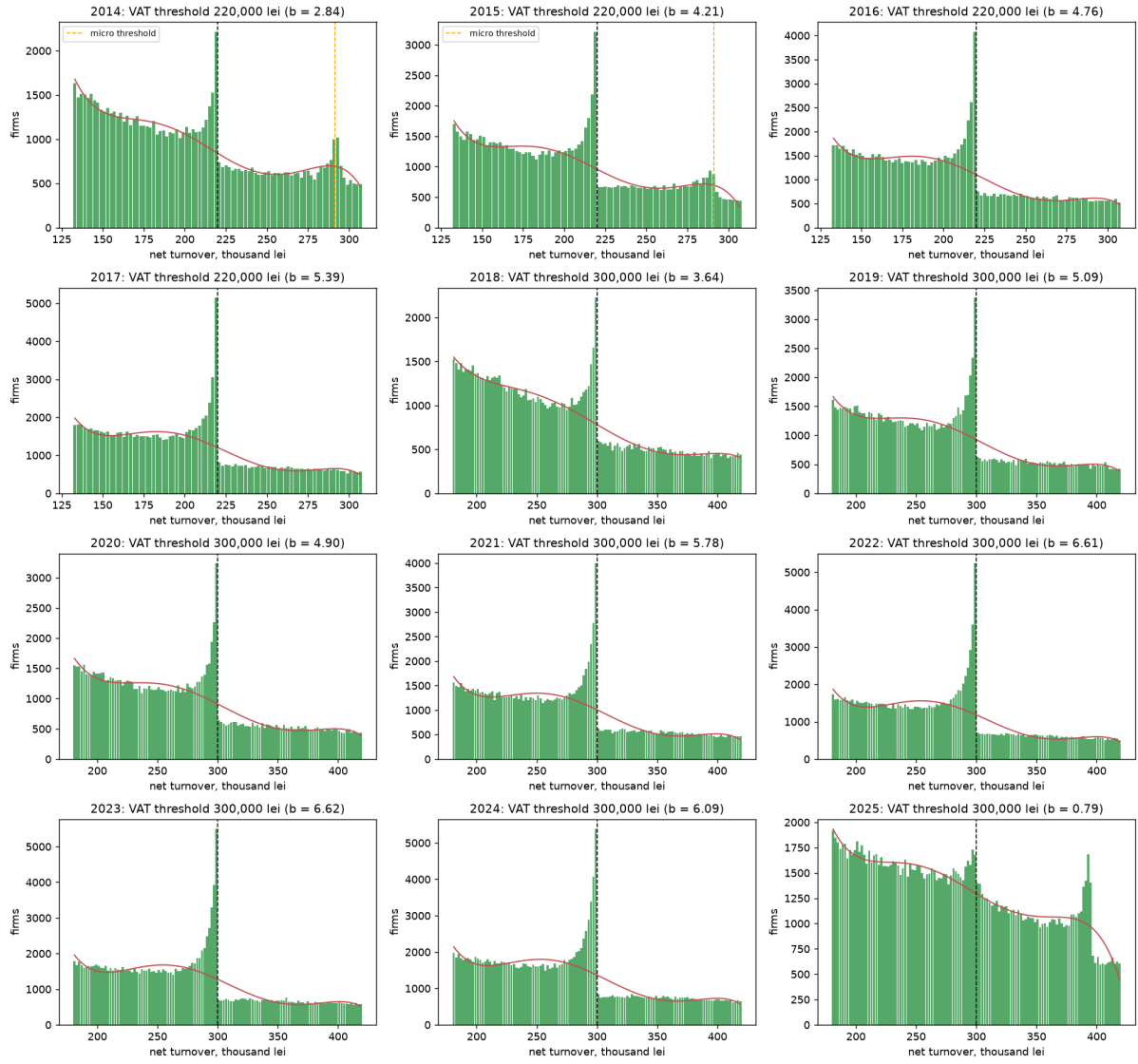


Figure 5: Net turnover around the VAT registration threshold, 2,000-lei bins, with a polynomial counterfactual.

Table 17: Sibling-firm indicators by position relative to the threshold, SRLs, pooled 2018–2023

Measure	Bunchers	Just above	10–25% below	25–100% above	z
Administrator registered another firm, prior 18 months	2.71	1.98	1.72	1.91	4.3
New firm at the same address, prior 18 months	12.23	9.18	8.72	8.37	8.6
Administrator runs another firm, any date	17.09	15.88	15.16	17.46	2.6

Pooled shares, %. z tests bunchers against firms just above the threshold with standard errors clustered by firm. At the small ceilings of 2015–2017 the corresponding shares for bunchers and firms just above are 1.59 versus 1.37 ($z = 1.5$) on the administrator measure, 7.17 versus 5.34 ($z = 6.3$) on the address measure and 12.42 versus 11.38 ($z = 2.7$) on the any-date measure. Standardising the just-above group to the bunchers’ age distribution gives 2.70 versus 2.20 on the administrator measure and 12.26 versus 10.23 on the address measure (Appendix Table 18). Administrators as recorded in the ONRC snapshot of 2 September 2026; persons administering more than 50 firms and addresses hosting more than 20 firms excluded. Bunchers: 0–5% below the threshold; just above: 0–10% above.

D Splitting

Table 17 compares limited-liability companies in the 5 percent below the ceiling with firms just above it, which are the same size, and with firms further below and well above, pooled over 2018–2023 with standard errors clustered by firm. Dropping administrators without a birth date gives 2.4 against 1.7 percent, the same ratio. The share whose administrator runs any other firm, regardless of date, differs little between bunchers and firms just above ($z = 2.6$), so the timing measures carry the information. Bunchers are younger than firms just above, 13 percent under three years old against 8, and Table 18 compares within age bands: bunchers exceed the just-above group in every band, and standardised to the bunchers’ age distribution the premium is 23 percent on the administrator measure and 20 percent on the address measure. Two checks follow from the 2024 rules, which limit a person to one micro-enterprise and test the ceiling on linked enterprises combined. The administrator premium disappears in 2024, 1.4 percent for both groups against 2.0 and 1.6 in 2023, while a gap remains on the address measure; but Table 19 shows no excess mass in the combined revenue of firms sharing an administrator at €500,000 in 2024, and firms inside such groups bunch individually at the same rate as single firms. Either the aggregation rule was not enforced or noticed in its first year, or common administration is too weak a proxy for the ownership links the rule is written on. Firms that lost the regime when the ceiling fell registered new firms at the same rate, relative to the band below, as placebo cohorts.

Table 18: Sibling-firm indicators by firm age, bunchers versus firms just above the threshold, pooled 2018–2023 (%)

Firm age (years)	Bunchers, n	Just above, n	Admin. new firm: bunchers	just above	Same-address new firm: bunchers	just above
0-2	2,166	1,482	6.97	5.47	26.13	20.51
3-5	2,544	2,332	3.26	2.57	18.20	16.51
6-10	3,662	3,826	2.48	2.12	13.68	11.53
11+	8,158	10,187	1.50	1.25	6.09	4.96

Age = year minus year of registration in the trade register. Standardising the just-above group to the bunchers' age distribution gives 2.70 versus 2.20 (administrator measure, +23%) and 12.26 versus 10.23 (address measure, +20%).

Table 19: Bunching of combined revenue of firms sharing an administrator, and firm-level bunching inside groups

Year	Groups (2–5 SRLs, one administrator)	Groups in window	b , combined revenue at €500k	Excess groups	Firms in groups: % in 5% below own notch	Single firms: same
2021	13,563	970	0.11	3	19.5	18.1
2022	15,156	1,118	0.58	16	18.9	17.5
2023	15,860	1,304	0.90	29	21.5	20.1
2024	17,323	1,417	0.07	3	16.2	16.9
2025	18,044	1,335	0.62	20	15.8	14.5

From 2024 the ceiling is tested on the combined revenue of linked enterprises. Combined revenue of each administrator's firms, window $\pm 20\%$ around €500,000 at the previous year-end rate, controls = same-quantile density of combined revenue in 2019–2022 excluding years where the individual €1M notch is within 25%. Last two columns: share of firms within 20% of that year's individual notch that sit in the 5% just below it.

E Additional tables

Table 20 lists the admissible control years for each notch with the matched revenue and distance to the nearest notch. Table 21 gives further robustness checks; three others are not tabulated: the excess counts vary across the 3, 5 and 8 percent region definitions by 6 to 50 percent, mostly at the two extra notches; dropping hospitality firms, which faced no ceiling in 2023, raises the 2023 estimate from 2.99 to 3.14, and hospitality firms alone show 0.58; newly created firms make up 11 percent of the 2023 excess, against 7 percent in 2021 and 13 percent in 2024. Table 22 gives the origin of firms just below €500,000 in the notch year and in placebo years. Table 23 gives the wide-band event study. Table 24 gives the reduced-form elasticities discussed in Section 4; in 2016 the median buncher paid 1 percent, not the 3 percent the table assumes.

Table 20: Admissible control years, matched locations and distance to the nearest notch

Notch	Control year: matched revenue L_c^* (distance to nearest notch in that year)
2014: in-year, €65,000	2016: 0.32M (32%); 2017: 0.33M (34%); 2022: 0.44M (31%); 2023: 0.46M (35%); 2024: 0.47M (36%)
2015: in-year, €65,000	2016: 0.30M (26%); 2017: 0.30M (27%); 2023: 0.43M (30%); 2024: 0.43M (31%)
2016: in-year, €100,000	2014: 0.42M (30%); 2015: 0.47M (38%); 2017: 0.48M (54%); 2018: 0.50M (40%); 2019: 0.52M (42%); 2020: 0.49M (39%); 2021: 0.57M (47%); 2022: 0.62M (52%); 2023: 0.66M (54%); 2024: 0.66M (54%)
2017: in-year, €500,000	2014: 2.09M (86%); 2015: 2.25M (87%); 2016: 2.25M (80%); 2018: 2.42M (88%); 2019: 2.50M (86%); 2020: 2.39M (88%); 2021: 2.80M (74%); 2025: 3.13M (60%)
2018: in-year, €1,000,000	2014: 4.19M (93%); 2015: 4.49M (94%); 2016: 4.44M (90%); 2017: 4.54M (50%); 2023: 5.83M (58%); 2024: 5.88M (58%); 2025: 6.26M (80%)
2019: in-year, €1,000,000	2014: 4.12M (93%); 2015: 4.41M (93%); 2016: 4.37M (90%); 2017: 4.46M (49%); 2023: 5.73M (57%); 2024: 5.79M (57%); 2025: 6.16M (80%)
2020: in-year, €1,000,000	2014: 4.42M (93%); 2015: 4.73M (94%); 2016: 4.69M (90%); 2017: 4.76M (52%); 2023: 6.15M (60%); 2024: 6.21M (60%); 2025: 6.60M (81%)
2021: in-year, €1,000,000	2014: 3.92M (93%); 2015: 4.19M (93%); 2016: 4.15M (89%); 2017: 4.24M (46%); 2023: 5.44M (54%); 2024: 5.48M (55%); 2025: 5.84M (79%)
2022: in-year, €1,000,000	2014: 3.56M (92%); 2015: 3.82M (92%); 2016: 3.79M (88%); 2017: 3.87M (41%); 2023: 4.94M (50%); 2024: 5.01M (50%); 2025: 5.32M (77%)
2023: in-year, €500,000	2014: 1.79M (84%); 2015: 1.94M (85%); 2016: 1.94M (77%); 2018: 2.09M (86%); 2019: 2.16M (86%); 2020: 2.07M (86%); 2021: 2.42M (88%); 2025: 2.70M (54%)
2024: in-year, €500,000	2014: 1.75M (83%); 2015: 1.91M (85%); 2016: 1.90M (76%); 2018: 2.05M (85%); 2019: 2.12M (86%); 2020: 2.03M (85%); 2021: 2.37M (87%); 2025: 2.65M (53%)
2025: in-year, €250,000	2014: 0.82M (64%); 2015: 0.90M (68%); 2016: 0.89M (49%); 2017: 0.92M (76%); 2018: 0.97M (69%); 2019: 1.01M (70%); 2020: 0.96M (69%); 2021: 1.11M (73%); 2022: 1.21M (75%); 2023: 1.25M (76%); 2024: 1.23M (76%)
2022: announced, €500,000	2014: 1.69M (83%); 2015: 1.84M (84%); 2016: 1.83M (75%); 2018: 1.97M (85%); 2019: 2.05M (85%); 2020: 1.96M (85%); 2021: 2.28M (87%); 2025: 2.56M (51%)
2025: year-end, €100,000	2016: 0.34M (31%); 2017: 0.35M (37%); 2021: 0.42M (29%); 2022: 0.46M (35%); 2023: 0.49M (39%); 2024: 0.50M (40%)

L_c^* = revenue at the notch's quantile in the control year. Distance = $|n/L_c^* - 1|$ for the nearest known notch n of that year (micro ceiling, future ceiling, €60,000 boundary or VAT threshold); admissible if above 25%. The closest cases are 26–34%.

Table 21: Further robustness checks of the difference-in-bunching estimates

Notch	b	Quantile-corrected	Pre-years only	Post-years only	Continuing firms	n continuing	Jackknife SE
2014: in-year, €65,000	1.15	1.14	–	1.15	1.15	31,696	0.04
2015: in-year, €65,000	1.71	1.70	–	1.71	1.71	30,382	0.02
2016: in-year, €100,000	1.56	1.56	1.30	1.63	1.45	29,322	0.08
2017: in-year, €500,000	0.76	0.77	0.62	0.85	0.72	16,321	0.05
2018: in-year, €1,000,000	1.41	1.36	1.44	1.37	1.35	11,010	0.04
2019: in-year, €1,000,000	1.67	1.75	1.64	1.70	1.58	12,513	0.06
2020: in-year, €1,000,000	1.75	1.87	1.64	1.90	1.67	11,535	0.09
2021: in-year, €1,000,000	2.15	2.19	2.23	2.04	2.08	13,668	0.08
2022: in-year, €1,000,000	1.92	1.97	1.82	2.05	1.77	14,756	0.09
2023: in-year, €500,000	2.99	2.99	3.02	2.82	2.80	26,100	0.05
2024: in-year, €500,000	1.62	1.65	1.64	1.52	1.58	24,967	0.04
2025: in-year, €250,000	0.73	0.72	0.73	–	0.61	35,470	0.04
2022: announced, €500,000	0.80	0.79	0.80	0.83	0.83	21,379	0.03
2025: year-end, €100,000	1.47	1.47	1.47	–	1.43	51,879	0.04

Quantile-corrected: control quantile shifted down by B/N_t . Pre/post: only control years before/after the treated year. Continuing: firms with positive revenue in $t - 1$, t and $t + 1$. Jackknife SE over control years.

Table 22: Origin of firms just below €500,000: share that was above €500,000 the year before

Year	Status	Firms 0–6% below	% above €500k in $t-1$	% in 120–200% band in $t-1$	Firms 14–20% below	% above in $t-1$	Ratio
2019	placebo	2,999	23.6	8.3	3,922	12.3	1.92
2020	placebo	2,895	40.7	17.5	3,817	25.0	1.63
2021	placebo	3,058	19.4	7.3	3,999	9.9	1.96
2023	notch year	6,646	29.6	15.1	5,166	16.6	1.78
2024	post-transition	5,659	21.7	7.3	5,403	14.0	1.55

Positions relative to €500,000 at the previous year-end rate in every year, whether or not it was a threshold. Ratio = bunchers' share from above divided by the comparison group's share.

Table 23: Reported pre-tax margins around regime switches induced by ceiling changes

Year	Micro share T / C (%)	Median margin T (%)	Median margin C (%)	DiD median (pp)	DiD mean (pp)	Median revenue T / C (k€)
<i>2018 increase: (500k, 1M] vs (1M, 2M] on 2017 revenue</i>						
2015	15 / 14	5.39	5.07	-0.32	-0.39	559 / 1,108
2016	14 / 14	5.80	5.41	-0.25	-0.43	628 / 1,232
2017	27 / 15	6.00	5.36	+0.00	+0.00	684 / 1,351
2018	72 / 18	6.37	4.93	+0.80	+0.85	728 / 1,438
2019	61 / 22	6.22	4.89	+0.69	+0.65	746 / 1,466
2020	58 / 26	6.07	5.06	+0.37	-0.08	675 / 1,342
<i>2023 cut: (500k, 1M] vs (1M, 2M] on 2022 revenue</i>						
2020	74 / 51	9.84	7.70	-0.70	-1.14	462 / 912
2021	75 / 48	11.87	9.36	-0.33	-0.51	576 / 1,090
2022	75 / 36	11.76	8.92	+0.00	+0.00	696 / 1,351
2023	55 / 24	8.40	6.69	-1.13	-1.49	676 / 1,358
2024	24 / 18	6.43	5.69	-2.10	-2.49	691 / 1,381
2025	18 / 17	5.80	5.35	-2.39	-3.22	686 / 1,366
<i>Placebo: same bands on 2019 revenue, no regime change</i>						
2017	46 / 24	7.35	5.66	-1.47	-1.60	533 / 1,099
2018	74 / 37	8.24	5.82	-0.74	-0.88	621 / 1,249
2019	71 / 26	8.95	5.79	+0.00	+0.00	692 / 1,364
2020	70 / 21	7.85	5.54	-0.85	-0.98	640 / 1,270
2021	66 / 29	9.43	6.63	-0.36	-1.17	718 / 1,421

Balanced panels of firms with positive revenue in every year shown. T = firms whose base-year revenue placed them in the band that changed regime; C = firms in the band above that did not. DiD = change in T minus change in C relative to the base year (2017, 2022, 2019). Margins are pre-tax profit over revenue, winsorised at ± 1 . The placebo shows the drift induced by selecting on base-year revenue alone.

Table 24: Reduced-form elasticities under the Kleven–Waseem approximation, reported for completeness

Notch	τ	median buncher margin	Δt (pp)	$\Delta z^*/z^*$ from b (%)	e from b	hole width (%)	e from hole
2014: in-year, €65,000	3%	0.160	-0.44	1.15	undefined (dt<=0)	–	undefined (dt<=0)
2015: in-year, €65,000	3%	0.169	-0.30	1.71	undefined (dt<=0)	20	undefined (dt<=0)
2016: in-year, €100,000	3%	0.134	-0.86	1.56	undefined (dt<=0)	19	undefined (dt<=0)
2017: in-year, €500,000	1%	0.089	+0.42	0.76	0.007	20	4.67
2018: in-year, €1,000,000	1%	0.106	+0.70	1.41	0.014	18	2.3
2019: in-year, €1,000,000	1%	0.117	+0.87	1.67	0.016	18	1.84
2020: in-year, €1,000,000	1%	0.132	+1.11	1.75	0.014	19	1.61
2021: in-year, €1,000,000	1%	0.160	+1.56	2.15	0.015	19	1.15
2022: in-year, €1,000,000	1%	0.166	+1.66	1.92	0.011	20	1.2
2023: in-year, €500,000	1%	0.200	+2.20	2.99	0.02	–	hole not identified
2024: in-year, €500,000	3%	0.151	-0.58	1.62	undefined (dt<=0)	19	undefined (dt<=0)
2025: in-year, €250,000	3%	0.135	-0.84	0.73	undefined (dt<=0)	20	undefined (dt<=0)
2022: announced, €500,000	1%	0.140	+1.24	0.80	0.003	18	1.29
2025: year-end, €100,000	3%	0.234	+0.74	1.47	0.014	20	2.61

$\Delta t = 0.16 \times \text{margin} - \tau$ is the change in the average tax rate on revenue at the notch for a firm with the median reported margin of bunchers; $e = (\Delta z^*/z^*)^2 / (2\Delta t / (1 - \tau))$. Two proxies for $\Delta z^*/z^*$: b with 1% bins, and the width above the notch at which the cumulative deficit offsets the excess (the hole), which is not reached before the window edge in most years. The proxies differ by three orders of magnitude and $\Delta t \leq 0$ in the 3% years, which is why the text does not rely on these numbers.

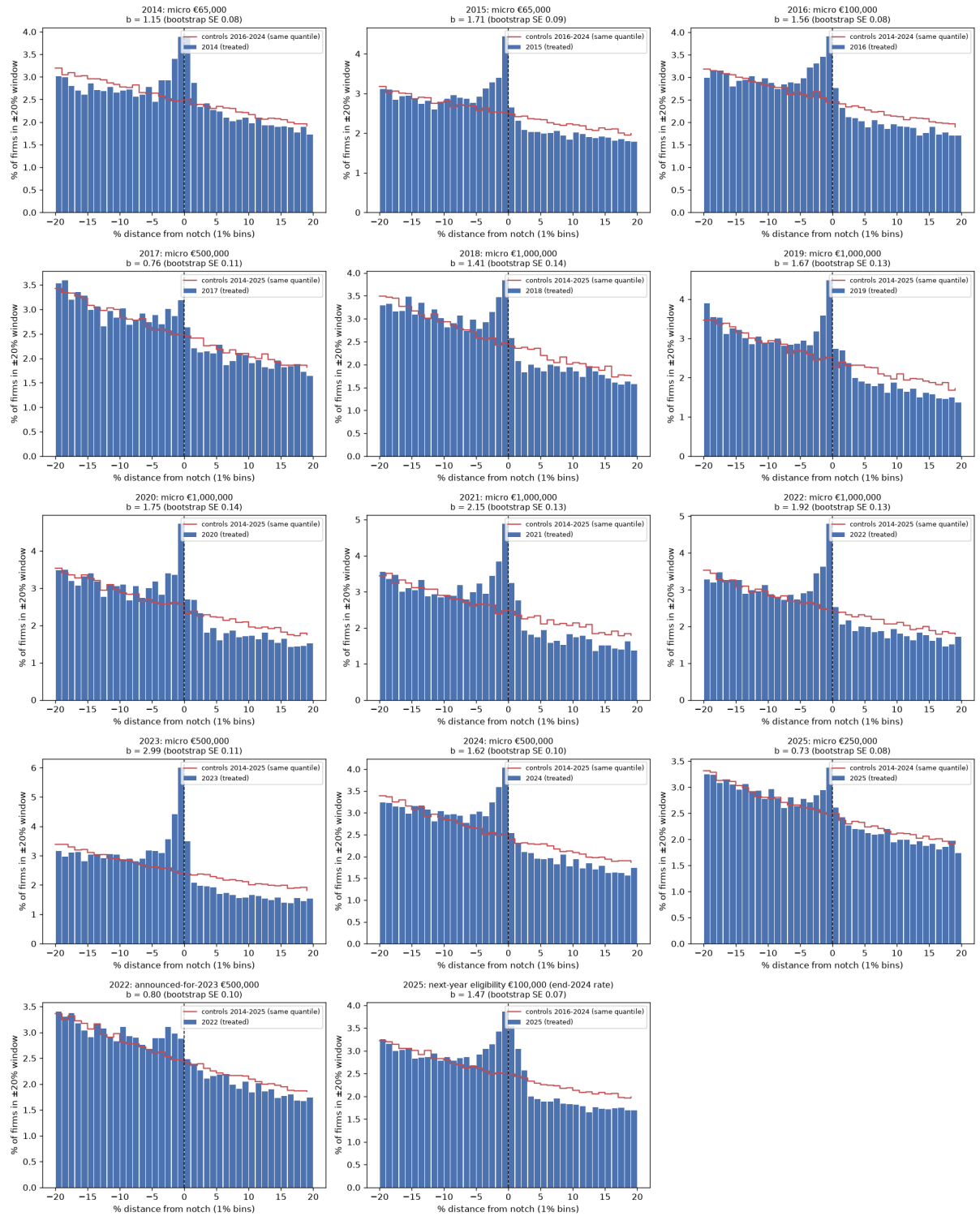


Figure 6: Treated-year density (bars) and same-quantile control density (line) around each of the fourteen notches, bins of 1 percent of the notch location.

F Additional figures

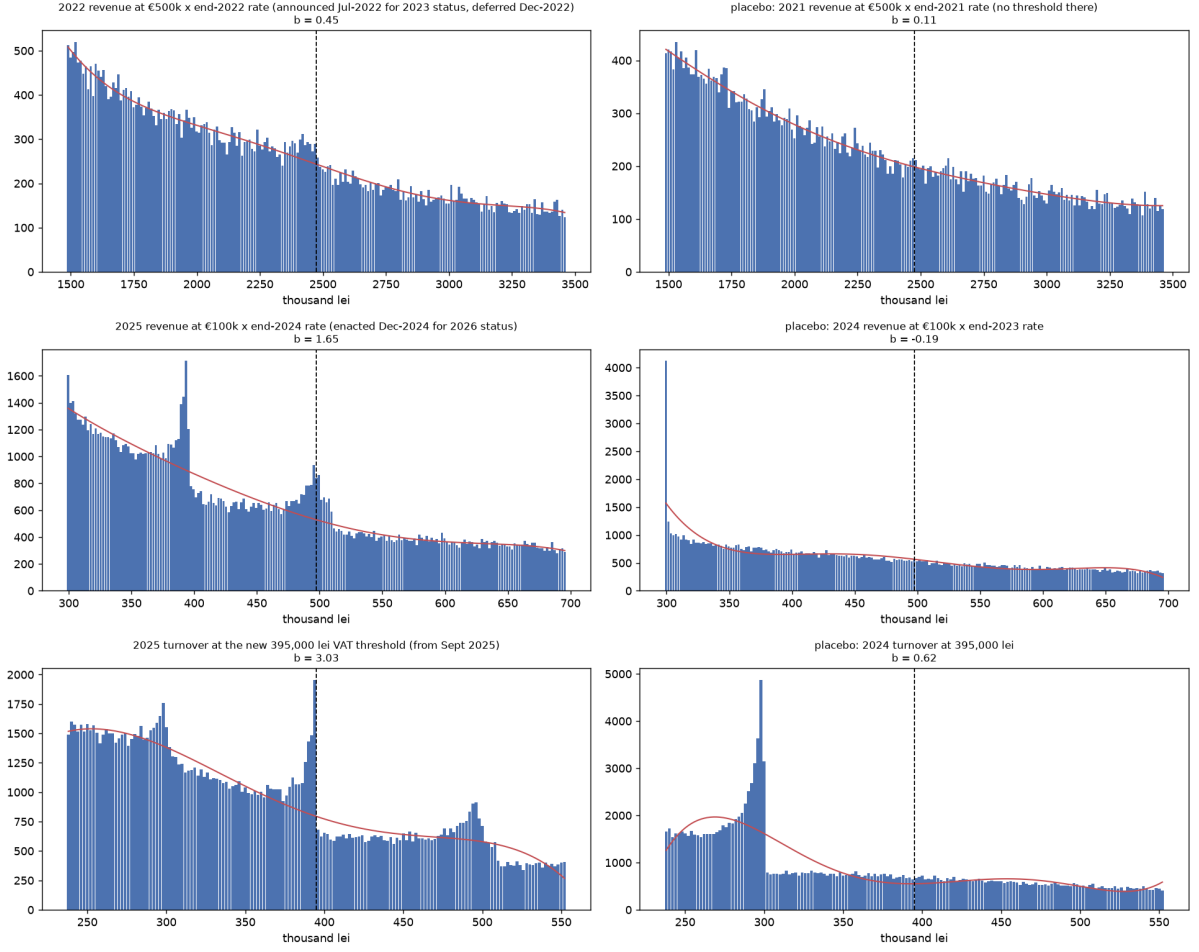


Figure 7: Revenue around announced or enacted future ceilings and the new VAT threshold, with placebo years. Top: 2022 revenue at €500,000, announced in July 2022 for 2023 status and deferred in December 2022, with 2021 placebo. Middle: 2025 revenue at €100,000 at the end-2024 rate, enacted December 2024 for 2026 status, with 2024 placebo. Bottom: 2025 turnover at 395,000 lei with 2024 placebo. Polynomial counterfactuals for reference.

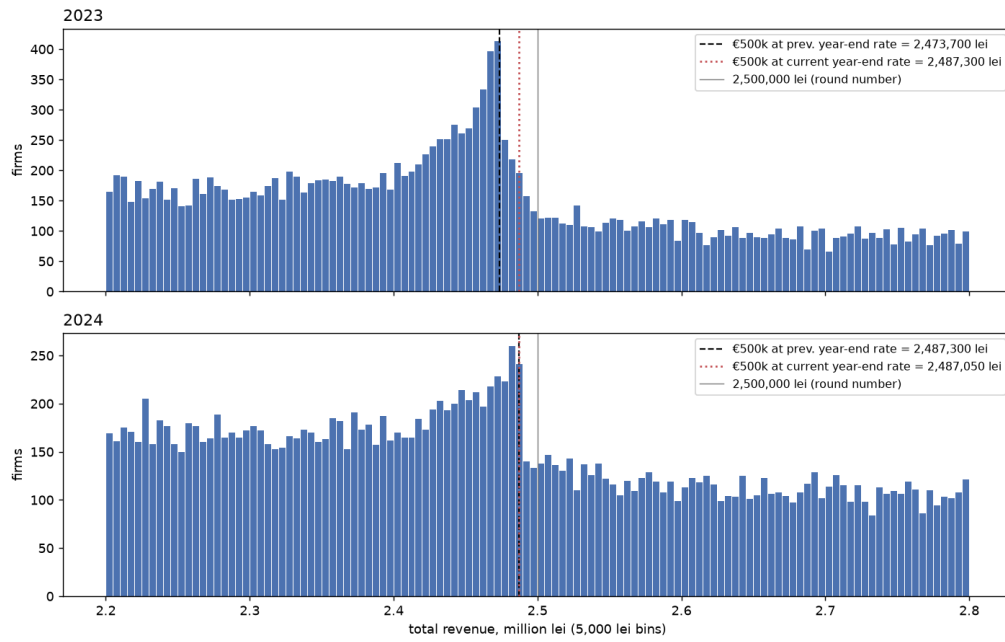


Figure 8: Total revenue in lei around €500,000, 2023 and 2024, 5,000-lei bins. The grey line marks 2,500,000 lei, the nearest round number.

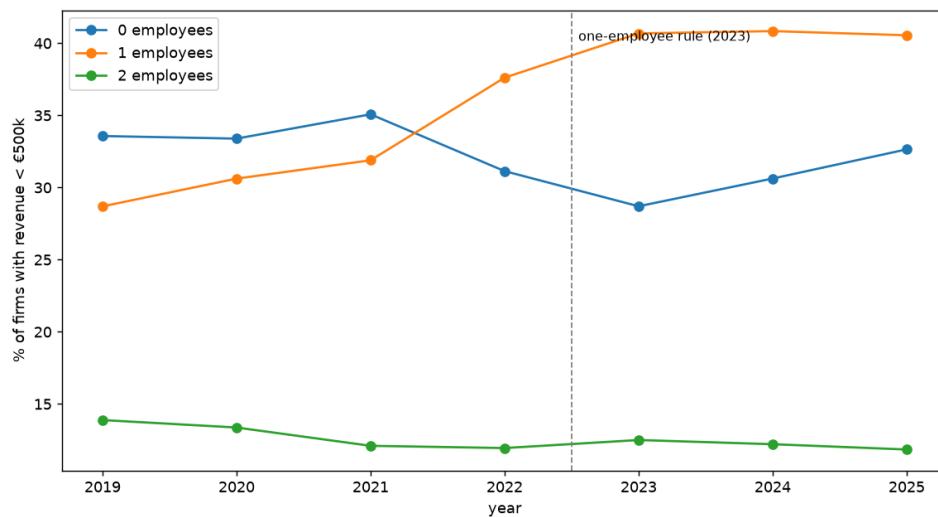


Figure 9: Share of firms with revenue below €500,000 reporting zero, one and two employees, 2019–2025.